

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA0202

COURSE OUTCOME COVERED

C03: *Analyze shape representation methods and techniques such as contours, regions, deformable models, and multi-resolution analysis. (BL4)*

Assessment Tool Weightage: 30%

QUESTIONS: 2 Total Marks:30 Annexure A
Case study

Code of Conduct:

*I _JAASIEL(192524046)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

		insights				
Soluti on Desig n	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justificatio n	Incorrect or no solution provided	
Clarit y	10	Clear, well structured, and easy to understand	Organized and understanda ble presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Annexure A

Case study

Case Study 1: Defect Detection in Industrial Parts

A manufacturing company wants to automatically detect defects in gears and machine components using computer vision. The images of the components are binary and may contain small protrusions or missing parts.

Questions:

- A. How can region-based representation and medial axis computation help in identifying defective parts?

Ans: Region-based representation considers the complete region occupied by the industrial component rather than only its outer boundary. For a binary gear or machine part, the object's area, connected regions, holes, and shape structure can be analyzed.

The medial axis gives a skeleton-like representation of the object by describing its central structure. For a normal component, the medial axis should have a consistent structure. A missing part, protrusion, or hole changes the skeleton and can therefore indicate a defect.

Thus, region properties together with the medial axis can be used to compare a test component with a normal reference component and identify structural abnormalities.

- B. Discuss how noise in image acquisition could affect shape analysis and propose preprocessing steps to improve accuracy.

Ans: Noise introduced during image acquisition can create small unwanted regions, false protrusions, broken boundaries, or artificial holes. These changes

can distort the region representation and medial axis, leading to incorrect defect detection.

Suitable preprocessing steps are:

1. Gaussian filtering to reduce random noise.
2. Median filtering to remove isolated pixels and small impulse noise.
3. Thresholding to obtain a clean binary image.
4. Morphological opening to remove small unwanted objects/protrusions.
5. Morphological closing to fill small gaps and holes.

These steps produce a cleaner binary shape and improve the reliability of subsequent shape analysis.

- c. Compare the effectiveness of Fourier descriptors versus wavelet descriptors for classifying normal vs. defective components.

Ans:

Feature	Fourier Descriptors	Wavelet Descriptors
Representation	Represents the contour in the frequency domain	Represents shape information at multiple resolutions
Main strength	Captures overall/global contour shape effectively	Captures both global shape and local details
Defect detection	Good for major contour changes	Better for localized defects such as small protrusions or missing portions
Multi-scale analysis	Limited compared with wavelets	Strong multi-resolution capability
Suitability	Effective when defects cause large boundary changes	More suitable when defects are small and localized

Case Study 2: Medical Image Segmentation

A hospital is developing a system to segment organs from CT scans. The organ boundaries are irregular and may vary slightly between patients.

Questions:

A. Explain how active contours (snakes) can be applied to segment the organ boundaries.

Ans: Active contours, or snakes, are deformable curves used to segment objects with irregular boundaries. An initial contour is placed around or near the organ in the CT image. The contour then iteratively deforms toward the organ boundary based on image information and contour constraints. The process stops when the contour reaches a stable boundary.

Thus, active contours can adapt to irregular organ shapes and provide a boundary for separating the organ from surrounding tissues.

B. Discuss the challenges of initializing contours and tuning parameters for accurate segmentation.

Ans: Accurate initialization is important because the initial contour must be placed sufficiently close to the organ boundary. If it is placed too far away or in the wrong region, the snake may converge to an incorrect boundary.

Parameter tuning is also challenging because the active contour depends on factors such as smoothness and image-edge attraction. Improper parameter values may cause the contour to become too smooth, fail to reach the organ boundary, or react to unwanted edges and noise. Therefore, a suitable initial contour and carefully selected parameters are necessary for accurate segmentation.

C. How could level set methods improve segmentation for cases where organs split or merge in the images?

Ans: Level set methods represent the evolving boundary using an implicit function and allow the boundary to change its topology during segmentation. Unlike a single fixed contour, the level set can naturally **split into multiple boundaries or merge into one boundary**.

Therefore, when organs appear to split or merge in medical images, level set methods can handle these changes more naturally and provide more flexible segmentation of complex anatomical structures.